

Volatility models in practice: Rough, Path-dependent or Markovian?

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Abstract

An extensive empirical study of the class of Volterra Bergomi models using SPX options data between 2011 and 2022 reveals the following fact-check on two fundamental claims echoed in the rough volatility literature:

Do rough volatility models with Hurst index $H \in (0, 1/2)$ really capture well SPX implied volatility surface with very few parameters? No, rough volatility models are inconsistent with the global shape of SPX smiles. They suffer from severe structural limitations imposed by the roughness component, with the Hurst parameter $H \in (0, 1/2)$ controlling the smile in a poor way. In particular, the SPX at-the-money skew is incompatible with the power-law shape generated by rough volatility models. The skew of rough volatility models increases too fast on the short end, and decays too slow on the longer end where “negative” H is sometimes needed.

Do rough volatility models really outperform consistently their classical Markovian counterparts? No, for short maturities they underperform their one-factor Markovian counterpart with the same number of parameters. For longer maturities, they do not systematically outperform the one-factor model and significantly underperform when compared to an under-parameterized two-factor Markovian model with only one additional calibratable parameter.

On the positive side: our study identifies a (non-rough) path-dependent Bergomi model and an under-parameterized two-factor Markovian Bergomi model that consistently outperform their rough counterpart in capturing SPX smiles between one week and three years with only 3 to 4 calibratable parameters.

JEL Classification: G13, C63, G10, C45.

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1 Introduction

1.1 Context

In the realm of (rough) volatility modeling, certain claims have gained widespread popularity and acceptance within academic circles and finance community. Specifically, it has been widely disseminated that rough volatility models exhibit exceptional performance, seemingly reproducing the stylized facts of option prices with remarkable precision while utilizing only a limited number of parameters. Furthermore, it has been argued that they outperform their traditional stochastic volatility model counterparts in capturing the essential features of volatility surfaces. These assertions have been made time and again in various research papers and were often presented with a high degree of confidence in contrast with little empirical evidence.

There is undoubtedly mathematical beauty in rough volatility¹, and “roughness is in the eye of the beholder” certainly sounds poetic. However as mathematicians, it is incumbent upon us to undertake a rigorous examination of these claims before echoing them. First, the assertions about the superior performance of rough volatility models appear to rest heavily on a limited set of visual fits confined to specific dates or time intervals, casting doubt on their applicability across a wider period. Furthermore, there has been notable omission in the literature concerning a *fair* and *comprehensive* comparison between rough volatility models and other models, such as the conventional Markovian stochastic volatility models and non-rough path-dependent volatility models. This absence leaves a critical gap in our understanding of the practical usefulness of rough volatility models, as the non-semimartingale and non-Markovian nature of rough volatility models have an important implementation cost that needs to be justified.

Recently, a series of independent empirical studies [7, 22, 38, 50], each focusing on different aspects of the volatility surface, presented evidence against the claim of “superior performance” attributed to rough volatility models in capturing the main characteristics of volatility surfaces. In particular, the following observations were made:

- The SPX At-The-Money (ATM) Skew does not follow a power law as prescribed in rough volatility literature [22, 38];
- Rough volatility models underperform, by a clear margin, their one-factor Markovian counterparts in all market conditions for short maturities with the same number of calibratable parameters for the joint SPX-VIX calibration problem [7];
- Rough volatility models suffer from severe structural limitations [7, 50].

Inspired by these studies, we conducted our own comprehensive empirical study using daily SPX implied volatility surface data from CBOE spanning from 2011 to 2022. Our study integrates various aspects explored in these earlier works and introduces new evidence and arguments to further challenge the perceived superiority of rough volatility models. We examine both the static performance of various models, such as their daily fit to the whole SPX volatility surface and to the ATM skew, as well as their dynamical performance by assessing how well they predict the future implied volatility surface. Moreover, our empirical study is divided into two parts: part one is dedicated to options with *short* maturities (one week to three months), with part two focusing on options with *long* maturities (one week to three years). This division offers additional insights into model performances at different timescales which could be useful in practice.

The empirical experiments in this paper are designed to ensure the comparison between various models is as fair as possible. To this end, we selected models from the same family of Volterra

¹From the theoretical side, there is no doubt that the rough volatility paradigm has been an inspiring source of motivation that lead to a better understanding of universal mathematical structures for a large class of non-Markovian and non-semimartingale Volterra processes, to cite a few [3, 4, 5, 6, 11, 15, 21, 29, 34, 40, 41, 43, 44, 51].

Bergomi models where the volatility process is driven by rough, path-dependent (non-rough semimartingale) and Markovian factors. The models we ended up with have (almost) the same number of calibratable parameters that can be interpreted in a similar way. Furthermore, the same daily forward variance curves extracted from market data are shared among all models for consistency.

As closed-form expression for option pricing under the family of Volterra Bergomi models are unavailable, we calibrate all models to market data using the generic unified numerical method ‘deep pricing with quantization hints’ developed in [7] based on functional quantization as a first proxy with Neural Networks (NN) as a corrector. Compared to other neural network structures used for pricing and calibration [42, 50], this method features significantly lower input dimensions (i.e. the forward variance curve of the Bergomi model is not part of the NN input). In addition, it is free from butterfly arbitrage by construction and mesh-free, allowing one to price derivatives for any strike and maturity combination without interpolation or extrapolation. Notably, this method also allows us to calibrate each model to the market ATM skew without relying on the approximation formula of the Bergomi-Guyon expansion [14] as in [22, 38]. This leads to a more accurate computation of the skew that is free from higher order error in the Bergomi-Guyon expansion when volatility of volatility (vol of vol) is large.

1.2 What are rough volatility models?

Rough volatility models are a specific subclass of non-Markovian stochastic volatility models, where the volatility process is assumed to be a non-semimartingale process characterized by continuous paths rougher than those of standard Brownian motion. Specifically, these models employ variants of fractional Brownian motion with a Hurst index less than $1/2$ to model the instantaneous volatility. For instance, in [10], the authors used a Riemann-Liouville fractional Brownian motion given by:

$$X_t = \int_0^t (t-s)^{H-1/2} dW_s, \quad (1.1)$$

where W is a standard Brownian motion to describe the spot volatility process. The Hurst index $H \in (0, 1/2)$ governs the local regularity of the path of X , indicating that X is Hölder continuous of order strictly less than H . In the special case of $H = 1/2$, the process X reduces to classical Brownian motion. The restriction $H > 0$ ensures that the convolution kernel $t \mapsto t^{H-1/2}$ is locally square-integrable so that the stochastic integral is well-defined as an Itô integral.

The Riemann-Liouville fractional Brownian motion (1.1) with $H \in (0, 1/2)$ is non-Markovian, as it represents a weighted average of the increments of Brownian motion by the kernel $t \mapsto t^{H-1/2}$. It is non-semimartingale due to the singularity of the kernel at $t = 0$, since $H < 1/2$. While historical volatility exhibits many path-dependent features such as volatility clustering, feedback and memory effects, and multiple timescales [18, 23, 39, 45, 47, 48], **the raison d’être of the rough volatility paradigm is to posit that high fluctuations and erratic/spiky behavior observed in volatility on short timescales arise from a rough continuous non-semimartingale process.**

To support the necessity of rough volatility and non-semimartingality, two empirical arguments were presented. First, the logarithm of realized volatility time series for various equity market indices exhibits statistical “rougher” trajectories (lower Hölder regularity) compared to standard Brownian motion [33]. An earlier but less popular empirical study of the roughness of the realized volatility appeared in [20] to motivate the need of a rough multi-fractional Brownian motion. Second, authors in [10, 17, 27, 31, 33] argued that market ATM skew term structure exhibits power law decay that explodes in the short term, a phenomenon that rough volatility models can reproduce [9, 10, 30].

However, the validity of both empirical arguments hinges on exceedingly fine timescales that are impractical to attain in finite data sets. Regarding the statistical roughness observed in the realized volatility time series, two independent studies [1, 49] as early as 2018 demonstrated that fast mean-

reverting continuous semi-martingales can produce spurious roughness at any realistic timescale, see also some more recent studies [19, 39] and Section 5.2 below. Recently, challenges to the claimed explosiveness of the short end of the ATM skew have been raised in [22, 38].

Setting aside the debate, one of our goals is to assess whether the inherent rough non-semimartingale component in rough volatility models, i.e. the explosiveness of the kernel $t \mapsto t^{H-1/2}$ at $t = 0$, aligns convincingly with market volatility surfaces across short and longer maturities as claimed in the rough volatility literature. It is crucial to emphasize that questioning a rough volatility model involves challenging its non-semimartingality, i.e. the explosion of the kernel at 0, rather than disputing its non-Markovian path-dependent nature. Said differently: are rough volatility models the ‘good’ non-Markovian models in practice?

1.3 Do rough volatility models really outperform significantly their classical Markovian counterparts?

This paper looks into two main aspects of model performance. First is the static aspect, measured by how accurate a model fits to the market smiles and ATM skew. Second is the dynamical aspect, measured by how well a model predicts future implied volatility.

Fitting short maturities between one week to three months: our empirical study shows that the rough Bergomi model actually underperforms on average compared to the one-factor Bergomi (Markovian) model with the same number of model parameters in fitting the volatility surface and the ATM skew. The one-factor Bergomi is also slightly better in predicting future volatility surface. In other words, there is clearly no advantage in using rough volatility models over classical Markovian models in the short term.

Fitting longer maturities between one week to three years: the rough Bergomi and the one-factor Bergomi models produce equally poor fits and are inconsistent with the SPX smiles, although the one-factor Bergomi model seems to at least be able to achieve a curved shape for the ATM skew in log-log plot more representative of the market, versus the straight line of rough volatility models. The rough Bergomi model slightly, but not consistently, outperforms the one-factor (Markovian) Bergomi model in fitting the volatility surface on average, but scores the highest variance. For the period between 2017 to 2019, the rough Bergomi model underperforms the one-factor Bergomi model. Furthermore, the rough Bergomi model significantly underperforms across all aspects compared to an under-parameterized two-factor (Markovian) Bergomi model with just one-extra calibratable parameter.

1.4 Do rough volatility models with Hurst index $H \in (0, 1/2)$ really capture SPX smiles with very few parameters?

Our study shows that rough volatility models are inconsistent with the global SPX volatility surface, caused by their severe structural limitations. In particular, the SPX ATM skew term structure cannot be modeled by the rigid power-law shape generated by rough volatility models. When one zooms in on the short end of the volatility surface, the SPX ATM skew does not increase as fast as that of rough volatility models, in line with [22, 38]. In addition, the power law decay $\sim T^{H-1/2}$ for long maturities observed on the market appears to be faster than $1/\sqrt{T}$ on average. Specifically, the average speed of decay of the ATM skew for large maturities over the last decade implies a negative value of H which is impossible for rough volatility models, since their H is constrained above zero.

On the other hand, we show that the path-dependent (non-rough) and the under-parameterized two-factor Bergomi model have no problem capturing the entire term structure of the SPX smiles and ATM skew via unconstrained $H \in (-\infty, 1/2)$ and can effectively decouple the short end

and the long end of the volatility surface in a parsimonious way, with only 3 to 4 calibratable parameters.

The superior performance of the non-rough path-dependent Bergomi model versus the rough Bergomi model highlights that the problem with rough volatility models seems to come from the non-semimartingale part, i.e. singularity of the kernel at 0. Indeed, the non-rough path-dependent model is obtained by smoothing out the singularity of the fractional kernel at 0 using a fixed value ε , which shows that the added flexibility of the path-dependent model comes from eliminating the rough, non-semimartingale behavior of rough volatility models and not from the calibration of an additional parameter. In volatility modelling, path-dependency is a desirable trait allowing a model to be more flexible at capturing several stylized facts of volatility, such as volatility clustering, feedback and memory effects, and multiple timescales. However, the non-semimartingality encoded in rough volatility models via the explosive kernel near $t = 0$ seems too restrictive and erodes the flexibility gained from path-dependency. Thus contrary to what is put forward by rough volatility advocates, roughness/non-semimartingality is not a necessary condition for effective volatility modelling. Furthermore, all models in this paper can generate spurious ‘statistical roughness’, see Section 5.2 which suggests that after all, estimation of the roughness of the realized volatility time series might not be sufficient to quantify the erratic and spiky behavior that characterizes the volatility of the market.

Outline of the paper: In Section 2, we will first introduce different Volterra Bergomi models considered in this paper and present their main relevant characteristics. In Section 3, we define the performance metrics used to evaluate model performance and address the specifics associated with model calibration. In Section 4, we present the empirical results on the static model performance in two parts, with part one dedicated to short maturities between one week to three months and part two for longer maturities between one week to three years. In the final Section 5, we compare the dynamical aspect of each model by looking at their prediction performance and estimating the statistical roughness of their simulated volatility paths.

2 The class of Volterra Bergomi models

The general form of the Volterra Bergomi stochastic volatility model for a (forward) stock price S with instantaneous variance V is defined as:

$$\begin{aligned} dS_t &= S_t \sqrt{V_t} dB_t, \\ V_t &= \xi_0(t) \exp \left(X_t - \frac{1}{2} \int_0^t K^2(s) ds \right), \\ X_t &= \int_0^t K(t-s) dW_s, \end{aligned} \tag{2.1}$$

with $B = \rho W + \sqrt{1 - \rho^2} W^\perp$ and $\rho \in [-1, 1]$. Here $(W, W^\perp)$ is a two-dimensional Brownian motion on a risk-neutral filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$. The process X is a centered Gaussian process with a non-negative locally square integrable kernel $K \in L^2([0, T], \mathbb{R}_+)$, in particular $X_t \sim \mathcal{N}(0, \int_0^t K^2(u) du)$, for all $t \leq T$. The deterministic input curve $\xi_0 \in L^2([0, T], \mathbb{R}_+)$ allows the model to match certain term structure of volatility (e.g. the forward variance curve):

$$\mathbb{E} \left[\int_0^t V_s ds \right] = \int_0^t \xi_0(s) ds, \quad t \geq 0.$$

The choice of the kernel K defines the dynamics of X (e.g. rough, path-dependent or Markovian) and plays a crucial role in the model’s capability of capturing the SPX (or other equity indices) volatility surface. In this paper, We shall consider the following kernels K and their corresponding model name in Table 1:

Model name	$K(t)$	Domain of H	Semi-mart.	Markovian
<i>rough</i>	$\eta t^{H-1/2}$	$(0, 1/2]$	✗	✗
<i>path-dependent</i>	$\eta(t + \varepsilon)^{H-1/2}$	$(-\infty, 1/2]$	✓	✗
<i>one-factor</i>	$\eta \varepsilon^{H-1/2} e^{-(1/2-H)\varepsilon^{-1}t}$	$(-\infty, 1/2]$	✓	✓
<i>two-factor</i>	$\eta \varepsilon^{H-1/2} e^{-(1/2-H)\varepsilon^{-1}t} + \eta_\ell \varepsilon^{H_\ell-1/2} e^{-(1/2-H_\ell)\varepsilon^{-1}t}$	$(-\infty, 1/2]$	✓	✓

Table 1: Different kernels K in this paper and their associated Bergomi model name. Note the two-factor Bergomi model means that the instantaneous variance process V is Markovian in two dimension driven by a single Brownian motion, see Section 2.4 for detailed explanations.

To ensure comparability among all models, we fix $\varepsilon = 1/52$ and $H_\ell = 0.45$. This means that all models contain the same number of calibratable parameters (η, ρ, H) , except the two-factor Bergomi model which takes on an additional calibratable parameter η_ℓ . The fixed values may seem arbitrary, however our goal is not about finding the optimal values for (ε, H_ℓ) , but rather to compare all models as fairly as possible. Furthermore, fixing (ε, H_ℓ) to these values does not alter the conclusion of this paper compared to setting these parameters free.

The particular parametrization of the one-factor and two-factor Bergomi models ensures that the parameter H has a similar interpretation across all these models. This would become more clear as we introduce each Bergomi model in the sections below.

We now introduce the ATM skew which is an important feature of the SPX volatility surface. The ATM skew $\mathcal{S}_T$ is defined as:

$$\mathcal{S}_T = \left. \frac{d\hat{\sigma}(T, k)}{dk} \right|_{k=0},$$

where $\sigma(T, k)$ is the implied volatility of vanilla options calculated by inverting the Black-Scholes formula with log-moneyness $k = \log(K/S_0)$ and maturity T . For the purpose of this paper, when we refer to the ATM skew, we would be actually referring to the negative value of $\mathcal{S}_T$.

2.1 The rough Bergomi

The rough Bergomi model of [10] defines X as:

$$X_t = \eta \int_0^t (t-s)^{H-1/2} dW_s,$$

with $\eta > 0$ the vol-of-vol parameter and $H \in (0, 1/2]$ the Hurst index that coincides with the roughness of the process X , i.e. the paths of X are Hölder-continuous of any order strictly less than H , almost surely. For $H < 1/2$, the fractional kernel $K(t) = t^{H-1/2}$ explodes as $t \rightarrow 0$, so that the process X is not a semi-martingale with trajectories rougher than that of standard Brownian motion. The restriction $H > 0$ ensures that the kernel K is locally square-integrable, so that the stochastic convolution is well-defined as an Itô integral.

The rough Bergomi model produces the following ATM skew (assuming flat ξ_0):

$$\mathcal{S}_T \approx \rho \eta T^{H-1/2} \quad (2.2)$$

at the first order of η , see [9, 10, 13, 30]. In particular, the skew explodes at $T \rightarrow 0$, with H controls both the skew explosive rate for small maturities and the rate of power-law decay for large maturities.

2.2 The path-dependent Bergomi

In the path-dependent Bergomi Model, the dynamics of X is governed by:

$$X_t = \eta \int_0^t (t + \varepsilon - s)^{H-1/2} dW_s. \quad (2.3)$$

The time shifted kernel, $K(t) = (t + \varepsilon)^{H-1/2}$, with $\varepsilon > 0$ has been independently introduced over the years in [2, 7, 13, 37]. It represents a small perturbation in the fractional kernel by $\varepsilon > 0$. However, this shift means $K(0)$ is finite, thus allowing the domain of $H \in (-\infty, 1/2]$ to be extended to $-\infty$ while K remains $L^2([0, T])$ integrable. The process X is now a semi-martingale and thus the sample paths have the same regularity as standard Brownian motion and are Hölder-continuous of any order strictly less than $1/2$.

The process X is not Markovian. To see this, we apply Itô's formula and get:

$$dX_t = \eta \left(\int_0^t (t + \varepsilon - s)^{H-3/2} dW_s \right) dt + \eta \varepsilon^{H-1/2} dW_t.$$

Next, with the help of the resolvent of the first kind of K , defined as the deterministic measure L such that $\int_0^t K(t-s)L(ds) = 1$, for all $t \geq 0$, with $L(dt) = \delta_0(dt)/K(0) + \ell(t)dt$, where ℓ is a completely monotone function, we get

$$dX_t = \left(-(1/2 - H)\varepsilon^{-1}X_t + \int_0^t f(t-s)X_s ds \right) dt + \eta \varepsilon^{H-1/2} dW_t, \quad (2.4)$$

with

$$f(t) = \frac{\varepsilon^{H-3/2}}{H-1/2} \ell(t) + \frac{1}{(H-1/2)(H-3/2)} \int_0^t (t + \varepsilon - u)^{H-5/2} \ell(u) du,$$

see [2, Lemma 1.2] for detailed similar computations. X is non-Markovian due to the part $\int_0^t f(t-s)X_s ds$ in the drift which depends on the whole trajectory of X up to time t , hence the name path-dependent.

The expression (2.4) provides essential insights to the dynamic of the process X : for small t , one expects the non-Markovian term $\int_0^t f(t-s)X_s ds$ to be negligible, so that the process X behaves locally like an Ornstein-Uhlenbeck process with large mean-reversion speed $(1/2 - H)\varepsilon^{-1}$ and vol of vol $\varepsilon^{H-1/2}$ for small ε . For large t , the non-Markovian term $\int_0^t f(t-s)X_s ds$ becomes more prominent and introduces path-dependency.

The path-dependent Bergomi model is very different to the rough Bergomi model. It is true that for $H \in (0, 1/2)$, by setting $\varepsilon = 0$ in (2.3), one recovers the rough Bergomi model. However for any $\varepsilon > 0$, the path-dependent Bergomi model produces a distinct model dynamic. This can be seen from its ATM skew formula [35] below in the first order of vol of vol, in contrast to that of the rough Bergomi model in (2.2):

$$S_T \approx \frac{\eta \rho}{2(H+1/2)T^2} \left(\frac{(T+\varepsilon)^{H+3/2}}{H+3/2} - \frac{\varepsilon^{H+3/2}}{H+3/2} - \varepsilon^{H+1/2} T \right).$$

This formula shows that the global shape of the ATM skew of the path-dependent Bergomi model is more flexible than the power-law shape of the rough Bergomi model. For very short maturities, the ATM skew of the path-dependent Bergomi model approaches a finite limit $\frac{\rho \eta \varepsilon^{H-1/2}}{4}$ which can be made as arbitrarily large as necessary via different values of ε , in contrast to the blow up to ∞ in the rough Bergomi model. For longer maturities, the ATM skew of the path-dependent Bergomi model decays at a rate $\sim \rho \eta T^{H-1/2}$, with the crucial difference that H can be negative, thus allowing the ATM skew to decay faster than that of rough Bergomi model.

For this paper, we fix the value $\varepsilon = 1/52$ without calibration. This choice not only ensures the same number of calibratable parameters as the rough Bergomi model, but also underscores that the flexibility of the path-dependent model comes from eliminating the rough, non-semimartingale behavior (i.e. removing the singularity of the kernel at $K(0)$) of rough volatility models and not from the calibration of an additional parameter, as we will see in Section 4.

2.3 The one-factor Bergomi

The one-factor Bergomi model introduced in [12] and [24] models X using a standard Ornstein Uhlenbeck process:

$$X_t = \eta \varepsilon^{H-1/2} \int_0^t e^{-(1/2-H)\varepsilon^{-1}(t-s)} dW_s,$$

where $H \in (-\infty, 1/2]$. For small ε and H , X has a large mean reversion speed of order $(1/2-H)\varepsilon^{-1}$ and a large vol of vol of order $\varepsilon^{H-1/2}$. This way of parameterizing X is reminiscent of models of fast regimes in [2, 4, 7, 26, 46] and can be seen as a Markovian proxy of the path-dependent Bergomi model from the previous section by dropping the non-Markovian term $\int_0^t f(t-s)X_s ds$ in (2.4).

This way of parametrization allows H to take on a similar interpretation to that in the path-dependent Bergomi model: the more negative the H , the statistically rougher the sample path of X driven by larger mean reversion and vol-of-vol, see Section 5.2 below. It also allows us to easily compare the models since they have the same calibratable parameters.

The ATM skew produced by the one-factor Bergomi model at first order of vol of vol, assuming a flat ξ_0 is of the form [14]:

$$\mathcal{S}_T \approx \frac{\rho}{2} \frac{\varepsilon^{H+1/2} \eta}{(1/2-H)T} \left(1 - \frac{(1 - e^{\frac{H-1/2}{\varepsilon} T}) \varepsilon}{(1/2-H)T} \right),$$

which shares the same finite limit $\frac{\rho \eta \varepsilon^{H-1/2}}{4}$ to that of the path-dependent Bergomi model. For small T , the ATM skew decay for both models are similar: a simple computation reveals that both models have the same limit in their first order derivative in T with respect to the skew:

$$\lim_{T \rightarrow 0} \frac{d\mathcal{S}_T}{dT} = \frac{\eta \rho (H - 1/2) \varepsilon^{H-3/2}}{12}$$

Therefore we can expect similar model behavior between the path-dependent and one-factor Bergomi models for very short maturities, see Section 4.1. On the other hand, the ATM skew decay of the one-factor Bergomi model is $\sim 1/T$ for large T .

2.4 The (under-parameterized) two-factor Bergomi

The two factor Bergomi model introduced in [12] contains two Ornstein Uhlenbeck factors X^1 and X^2 in the instantaneous variance V :

$$\begin{cases} X_t = X_t^1 + X_t^2, \\ X_t^1 = \eta \varepsilon^{H-1/2} \int_0^t e^{-(1/2-H)\varepsilon^{-1}(t-s)} dW_s, \\ X_t^2 = \eta_\ell \varepsilon^{H_\ell-1/2} \int_0^t e^{-(1/2-H_\ell)\varepsilon^{-1}(t-s)} dW_s, \end{cases}$$

where the parametrization of both factors are derived in the same way as the one-factor Bergomi model above with $(H, H_\ell) \in (-\infty, 1/2]^2$.

In the literature, the two factors are usually driven by two correlated Brownian motions. This would require two extra parameters to model the correlation between the factors themselves and

with the Brownian motion B in the spot process S in (2.1). In practice, the calibrated correlation between X^1 and X^2 are often very high, and for the sake of comparability and fairness among the models, we will use the same Brownian motion W for both factors, i.e. V is Markovian in two-dimension (X^1, X^2) driven by a single Brownian motion W .

By fixing $\varepsilon = 1/52$, $H_\ell = 0.45$, we induce a fast factor X^1 (with large mean-reversion and large vol of vol for small H) and a slow factor X^2 to mimic different scaling of volatility similar to the path-dependent Bergomi model without sacrificing the Markovian property. On the ATM skew, the two factors can decouple the short and long end of the term structure, with the fast factor exerting more influence on the short end and the slow factor becoming more important as T increases. Indeed, the ATM skew assuming a flat ξ_0 at first order of vol of vol of the two-factor (or any N factor) Bergomi model is linear in the contribution of each individual factor to the ATM skew [14]:

$$\mathcal{S}_T \approx \frac{\rho}{2} \left(\frac{\varepsilon^{H+1/2} \eta}{(1/2 - H)T} \left(1 - \frac{(1 - e^{\frac{H-1/2}{\varepsilon} T}) \varepsilon}{(1/2 - H)T} \right) + \frac{\varepsilon^{H_\ell+1/2} \eta_\ell}{(1/2 - H_\ell)T} \left(1 - \frac{(1 - e^{\frac{H_\ell-1/2}{\varepsilon} T}) \varepsilon}{(1/2 - H_\ell)T} \right) \right),$$

where $\mathcal{S}_T$ has a finite limit $\frac{\rho(\eta\varepsilon^{H-1/2} + \eta_\ell\varepsilon^{H_\ell-1/2})}{4}$ when sending $T \rightarrow 0$. For maturities up to three years, the two factor Bergomi models can mimic $\sim T^{H-1/2}$ power law decay [13, 35], despite of having the same asymptotic for very large T as the one-factor Bergomi model.

3 Model Assessment

Our empirical study involves calibrating each model described in Section 2 to the daily SPX volatility surfaces between August 2011 to September 2022 with market data purchased from the CBOE website, <https://datashop.cboe.com/>. In total, there are 2,807 days of SPX implied volatility surfaces.

To date, there are no closed form formulae for pricing vanilla options for the family of Volterra Bergomi models. To speed up the tedious numerical optimization procedure and to ensure a fair comparison, we rely on the generic-unified method ‘deep pricing with quantization hints’ developed in [7]. This method allows us to price vanilla derivatives efficiently and accurately by combining functional quantization and Neural Networks. In a nutshell, functional quantization is first used to obtain a decent approximation of the vanilla option price, with Neural Network trained on Monte Carlo prices added on later as a corrector. This approach is versatile enough to be applied to all models where volatility is a function of Gaussian process. The main advantages of this method are: 1) it is free from butterfly arbitrage by construction and mesh-free, allowing one to price any strike and maturity combination without interpolation or extrapolation, and 2) lower Neural Network input dimension allowing generalization over a wide range of forward variance curve $\xi_0(\cdot)$. Please refer to [7, Section 5] for detailed implementation.

3.1 Treatment of the forward variance curve

We use the same forward variance curve $\xi_0(\cdot)$ across all four models inferred directly from CBOE option prices via the well-known log-contract replication formula [16]. We further assume that $\xi_0(\cdot)$ is a piece-wise constant càdlàg function as suggested by Lorenzo Bergomi himself in [13]:

$$\xi_0(t) = \sum_{i=1}^N \mathbf{1}_{t \in [T_i, T_{i+1})} \xi_i,$$

where T_i are available SPX option maturities, $T_0 := 0$ and $\xi_i > 0$. We can extract ξ_i via:

$$(T_{i+1} - T_i)\xi_i = 2 \left(\int_0^{S_0} \frac{P(K, T_{i+1})}{K^2} dK + \int_{S_0}^{+\infty} \frac{C(K, T_{i+1})}{K^2} dK \right) - 2 \left(\int_0^{S_0} \frac{P(K, T_i)}{K^2} dK + \int_{S_0}^{+\infty} \frac{C(K, T_i)}{K^2} dK \right), \quad (3.1)$$

where $C(K, T_i)$ and $P(K, T_i)$ are the market prices of vanilla call/put options with strike K and maturity T_i . Due to the scarcity of market price for deep out of money options especially for negative log moneyness, we interpolated each slice of SPX smile using a pre-determined methodology (e.g. SSVI) then proceed with the computation in (3.1). Note we are only using the interpolated surface to estimate ξ_0 , the actual calibration is performed using the CBOE data.

3.2 Calibration performance metrics

For model evaluation, we investigate both the static performance of the model measured by how accurate it fits the global SPX smiles and the ATM skew, and the dynamical performance of the model, measured by how well it predicts future SPX smiles.

3.2.1 Global fit of the implied volatility surface

To test the global fit of the SPX smiles, each model is calibrated by minimizing the error function between model implied volatility surface and that of SPX over the set of calibratable model parameters, denoted collectively as Θ . For the rough, path-dependent and one-factor Bergomi models, $\Theta = \{\eta, \rho, H\}$. For the two-factor Bergomi model, $\Theta = \{\eta, \rho, H, \eta_\ell\}$. We chose the Root Mean Square Error (RMSE) as the error function $\mathcal{J}(\Theta)$:

$$\mathcal{J}(\Theta) := \sqrt{\frac{1}{|\mathcal{I}|} \sum_{(i,j) \in \mathcal{I}} \left(\hat{\sigma}_{i,j}^{mid} - \hat{\sigma}_{i,j}(\Theta) \right)^2}, \quad (3.2)$$

with $\hat{\sigma}_{i,j}^{mid}$ the SPX mid implied volatility with maturity T_i and strike K_j , and $\hat{\sigma}_{i,j}(\Theta)$ the model implied volatility. The set of available SPX implied volatility data for different maturities and strikes is captured by the index set $\mathcal{I}$, with $|\mathcal{I}|$ denoting the total number of available data points.

Due to the availability of market data as well as the stability of the implied volatility estimator, we use the following log-moneyness ($k = \log(K/S_0)$) range shown in Table 2 for model calibration:

Maturities T	log moneyness range
< 2 weeks	$[-0.15, 0.03]$
< 1 month	$[-0.25, 0.03]$
< 2 months	$[-0.3, 0.04]$
< 3 months	$[-0.4, 0.15]$
< 6 months	$[-0.6, 0.15]$
< 1 year	$[-0.8, 0.2]$
≥ 1 year	$[-1.5, 0.3]$

Table 2: Log moneyness range for different maturities considered for model calibration to the global SPX volatility surface.

3.2.2 Fit of the implied volatility ATM skew

To evaluate model fit of the ATM skew, each model is calibrated to the SPX ATM skew by minimizing the error between model implied volatility and the mid SPX implied volatility over a narrow range of log moneyness $k \in [-0.05, 0.03]$ across all maturities, as well as the error between log of model ATM skew and log of SPX ATM skew:

$$\min_{\Theta} \left\{ \sqrt{\frac{1}{|\mathcal{I}|} \sum_{(i,j) \in \mathcal{I}} \left(\hat{\sigma}_{i,j}^{mid} - \hat{\sigma}_{i,j}(\Theta) \right)^2 + \frac{1}{|\mathcal{M}|} \sum_{(m) \in \mathcal{M}} \left(\log \hat{S}_{T_m}^{mkt} - \log \hat{S}_{T_m}(\Theta) \right)^2} \right\},$$

where $\hat{S}_{T_m}^{mkt}$ is the SPX ATM skew at maturity T_m , with $\hat{S}_{T_m}(\Theta)$ is the model ATM skew. The set of available market ATM skew for different maturities is captured by the index set $\mathcal{M}$.

Thanks to the mesh-free nature of our deep pricing method that combines Functional Quantization and NN, we are able to calibrate each model directly to the SPX ATM skew without resorting to the approximation formula of the Bergomi-Guyon expansion as is the case in [22, 38]. While the Bergomi-Guyon expansion provides a fast approximation of model ATM skew, it can be error-prone in the case involving rough volatility [10] and situations where the vol of vol is large in Markovian-factor models.

To compute the ATM skew of each model, we first compute the model implied volatility near the money and then use the central finite difference at $k = 0$. The SPX ATM skew is computed by fitting a Lagrange polynomial of order 3 locally near the money and then taking the first order derivative at $k = 0$. We checked our results to ensure no over/under-fitting.

3.2.3 Prediction quality (parameter stability)

This performance metric tests the dynamical performance of each model by looking at their prediction quality. The main idea is that a robust parametric model should not require frequent re-calibration, since the change in the state variables (i.e. $\xi_0(\cdot)$) alone should be sufficient in justifying the change in the output (i.e. deformation in the implied volatility surface). Thus this performance metric can also be seen as a test on the stability of calibrated parameters.

To evaluate the prediction quality of a model, we perform the following experiment: for each trading day, we take the calibrated parameters Θ^* obtained by minimizing the error function (3.2) and keep it fixed for the next 20 working days. Next, we take the daily market forward variance curve $\xi_0(\cdot)$ extracted as per (3.1) and compute the error function (3.2) for each of the next 20 working days using Θ^* .

This is similar to the test described in Section 4.2 from [50]. However, the treatment of $\xi_0(\cdot)$ in [50] is not consistent for every model: the $\xi_0(\cdot)$ for the Markovian Heston model is modelled by only three parameters (V_0, κ, V_∞) , where the rough Bergomi and rough Heston models received preferential treatment by employing piece-wise constant function between maturities that offers greater flexibility in controlling the overall level of the model implied volatility. In addition, $\xi_0(\cdot)$ in [50] is re-calibrated daily for each model. In our experiment, we do not modify the market $\xi_0(\cdot)$ to ensure the state variable is being respected as much as possible.

4 Empirical results: the static comparison

This section on empirical performance is concentrated on the static performance of each model and is divided into two parts. The first part concentrate on model performance over the short term with maturities between one week to three months. The second part is dedicated to a larger range

of maturities spanning between one week to three years. We will defer the discussion on dynamical performance of each model to Section 5.

4.1 Short maturities (one week to three months)

We compare the calibration performance between rough, path-dependent and one-factor Bergomi models that share the same number of calibratable parameters. Figure 1 shows the evolution of daily calibration RMSE (3.2) of global fits of the implied volatility surface by each model. For ease of comparison, we show the monthly rolling average RMSE. The calibrated error for the path-dependent Bergomi model is almost always below that of the rough Bergomi model, while the one-factor Bergomi model also outperforms the rough Bergomi model about 63 percent of the time. The summary of statistics in Table 3 shows that the rough Bergomi model also scores the highest variance and the worst performance across the board. Some example fits are provided in Appendix A.

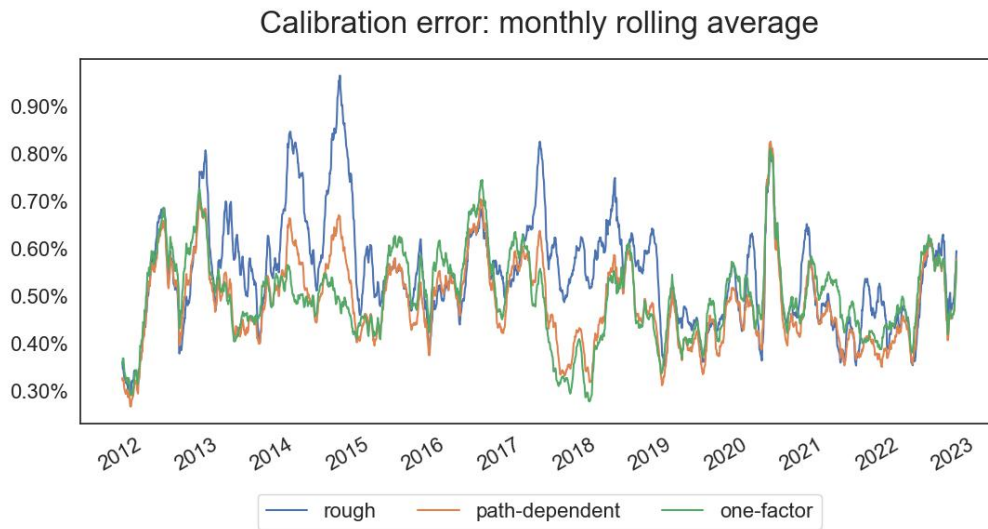


Figure 1: Evolution of monthly rolling average of calibration RMSE between different Bergomi models for maturities between one week to three months. All models share the same number of calibratable parameters, namely (η, ρ, H) .

Model	mean	std	min	5%	50%	95%	max
rough	0.005483	0.001786	0.001059	0.002996	0.005277	0.008661	0.01984
path-dependent	0.004859	0.001550	0.000937	0.002739	0.004697	0.007523	0.018898
one-factor	0.004987	0.001518	0.001070	0.002844	0.004811	0.007615	0.016218

Table 3: Statistics on the calibration RMSE for maturities between one week to three months. The lowest error for each statistical measure is in **bold**.

Figure 2 shows different model fits to the SPX ATM skew by plotting the average ATM skew and log of average ATM skew. The calibrated ATM skew from the one-factor Bergomi model is the closest to the market data for maturities up to one and half months, followed by the path-dependent Bergomi model. Even though no model seems to be able to capture perfectly the entire term structure of the ATM skew under three months, the rough Bergomi model failed to capture the market ATM skew at any point, characterized by an overly-explosive ATM skew near one week and a much too rapid decay straight after. We thus have evidence that rough volatility models are

inconsistent with market data at short timescales between one week and three months, and that ATM skew cannot be adequately explained by the power law generated by rough volatility models.

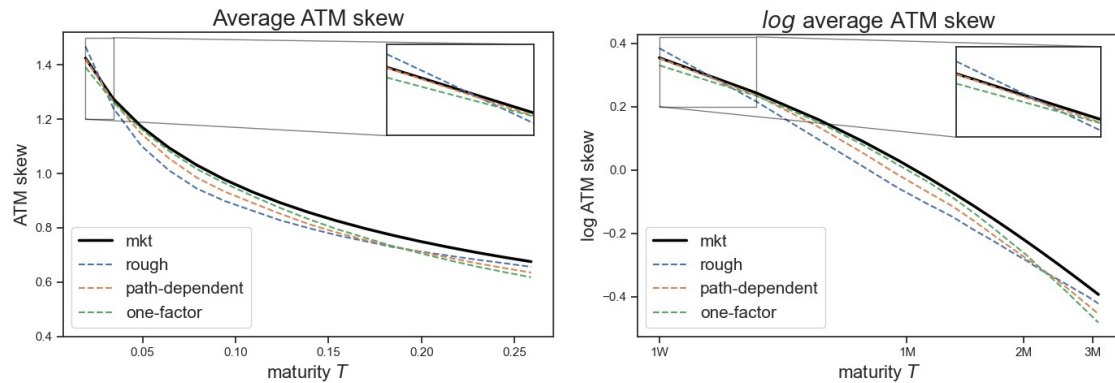


Figure 2: Average SPX ATM skew term structure (left) and log average SPX ATM skew term structure in log-scale (right) produced by different Bergomi models with maturities calibrated between one week to three months. We zoomed in for maturities between one to two weeks.

Contrary to what rough volatility literature claims, **we found no evidence supporting that rough volatility models fit better than their Markovian counterparts for short maturities of the SPX smile**. Instead, our study shows that the one-factor Bergomi actually outperforms the rough Bergomi model, which is in line with our earlier paper on joint SPX-VIX calibration [7, 8] where the Quintic one-factor Markovian model is shown to consistently outperform its rough counterpart for maturities up to 3 months.

Since all these models use the same calibratable parameters (η, ρ, H) , we can compare their evolution in time. Figure 3 shows the calibrated ρ for the rough Bergomi model tends to saturate near -1 (and even more so in Section 4.2 for longer maturities): this is a known structural issue of the rough Bergomi model, see [7, 25, 42, 50]. This saturation cannot be explained by the under-parametrization of the rough Bergomi model (since all the models here have the same number of parameters), but rather by its inconsistency with the market observation. Indeed, the other two models have very similar behavior in their calibrated parameters, which highlights again that the path-dependent model is very different from the rough model. Figure 4 shows that the calibrated H for the path-dependent and one-factor Bergomi models are almost always negative, versus that of the rough Bergomi model. The negative $H \leq -1/2$ can be related to jump regimes at the limit when $\varepsilon \rightarrow 0$, under Heston's dynamics [2].

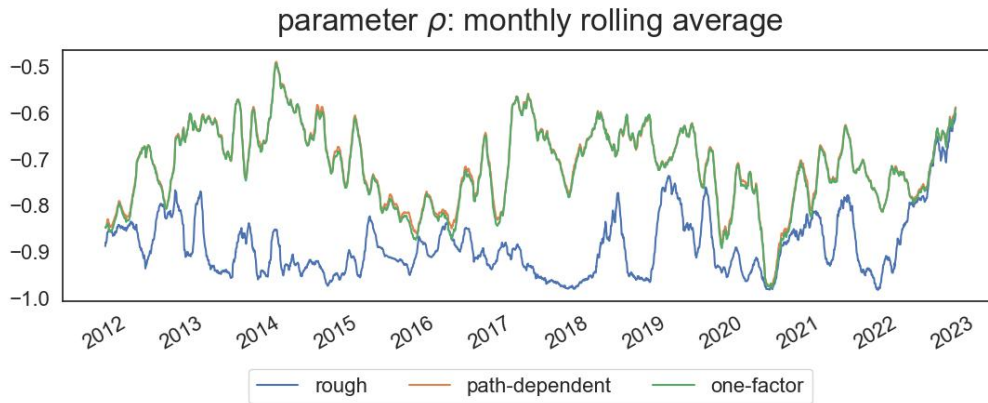


Figure 3: Evolution of monthly average of calibrated parameter ρ for different models calibrated to maturities between one week to three months.

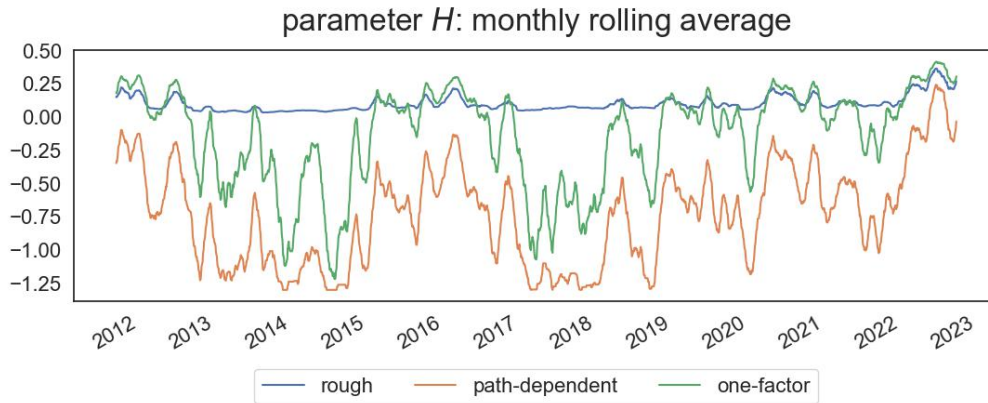


Figure 4: Evolution of monthly average of calibrated parameter H for different models calibrated to maturities between one week to three months.

4.2 Long maturities (one week to three years)

We now compare the empirical results for maturities running all the way to three years for the rough, path-dependent, one-factor and under-parameterized two-factor Bergomi models. We recall the two-factor Bergomi model contains one-extra calibratable parameter η_ℓ compared to the other models but with both factors driven by the same Brownian motion.

Figure 5 and Table 4 show the two-factor Bergomi model outperforms all other models for the global fit of the implied volatility surface, followed by the path-dependent Bergomi model. The performance of the rough Bergomi model is noticeably unsatisfactory. It scores again the highest variance. Even though its performance is slightly better than that of the one-factor Bergomi model (around 57.4 percent of the time), for the period between 2017 to 2019, it consistently underperformed the one-factor Bergomi model. Some example fits are provided in Appendix B.

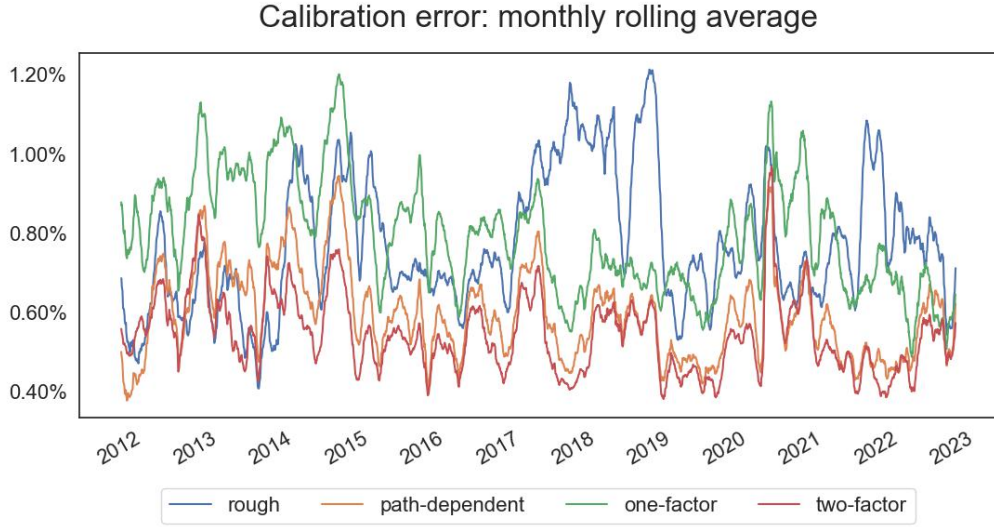


Figure 5: Evolution of monthly rolling average of calibration RMSE between different Bergomi models for maturities between one week to three years. Recall rough, path-dependent and one-factor Bergomi models share the same number of calibratable parameters (η, ρ, H) while the two-factor has an extra calibratable parameter η_ℓ .

Model	mean	std	min	5%	50%	95%	max
rough	0.007725	0.002113	0.002904	0.004755	0.007394	0.011517	0.020934
path-dependent	0.005935	0.001613	0.002271	0.003695	0.005755	0.008729	0.021469
one-factor	0.007943	0.001905	0.003287	0.005133	0.00781	0.011234	0.022543
two-factor	0.005431	0.001429	0.002403	0.003554	0.005272	0.007766	0.021952

Table 4: Statistics on the calibration error for the maturities between one week to three years. The lowest error for each statistical measure is in **bold**.

From Figure 6, the two-factor and the path-dependent Bergomi models produce on average a decent fit, although not perfect to the full term structure of the SPX ATM skew. Both the rough Bergomi and the one-factor Bergomi models produce equally bad fit and are inconsistent with the general shape of SPX ATM skew, although the one-factor Bergomi model seems to at least be able to achieve a curved shape in log-log plot more representative of the market vs. the straight line of rough volatility models. Like [22, 38], we now have further evidence challenging the assumption used by rough volatility advocates that the SPX ATM skew follows a power law. The straight line fit of the rough Bergomi model in the log-log plot on the right hand side is in stark contrast with the curved shape of market average log ATM skew term structure. In addition, its explosive behavior for small maturities results in overestimation for the short end; and for the long end the rough Bergomi model is unable to match the speed of decay of the market.

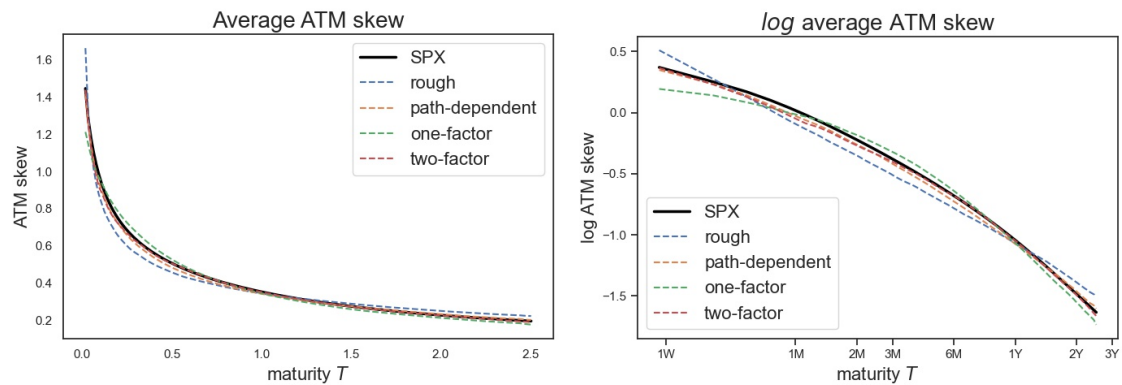


Figure 6: Average SPX ATM skew term structure (left) and log average SPX ATM skew term structure in log-scale (right) produced by different Bergomi models for maturities between one week to three years. Recall rough, path-dependent and one-factor Bergomi models share the same number of calibratable parameters (η, ρ, H) while the two-factor has an extra calibratable parameter η_ℓ .

Our results provides **strong evidence against the claim that rough volatility models consistently outperform their classical Markovian counterparts**. First, it is not clear that the rough Bergomi model actually performs better than the one factor Bergomi model. Second, the under-parameterized two-factor Bergomi model consistently outperforms the rough Bergomi model only with one additional parameter η_ℓ . As observed in [22, 50, 36], having two rough factors driving the volatility process in the Bergomi model does not significantly improve the model fit compared to the standard two factor (Markovian) Bergomi model.

Figure 7 shows that the calibrated ρ for the rough Bergomi model tends to saturate at -1 , making it somewhat a redundant parameter. The calibrated ρ for all other models are not saturated and move similar to one another:

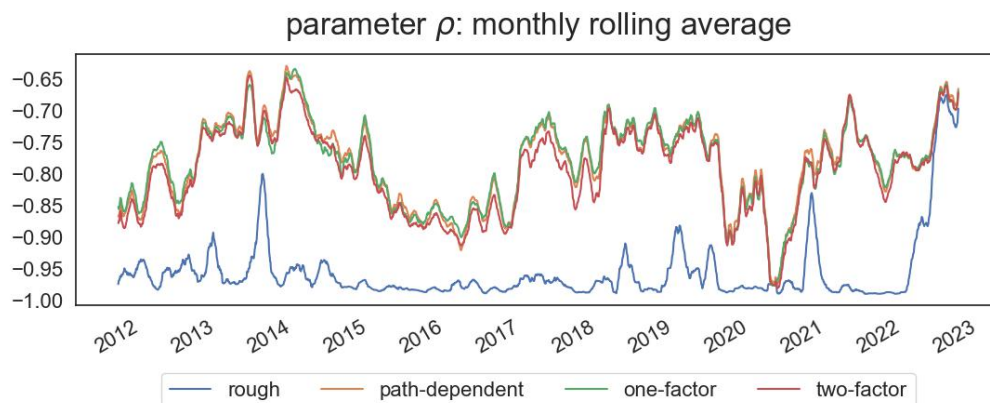


Figure 7: Evolution of monthly average of calibrated parameter ρ of different Bergomi models calibrated to maturities between one week to three years.

The calibrated H shown in Figure 8 for the path-dependent and two-factor Bergomi models are negative, in contrast with that of the rough Bergomi model. The H from the one-factor Bergomi model is usually the highest, but dipped below zero regularly for the period between 2017 and

2019. This coincides with the same time-frame where the one-factor Bergomi model outperformed the rough Bergomi model shown in Figure 5. Compared to Figure 4, the calibrated H here is about 0.4 higher for the path-dependent Bergomi model: the increase in H is necessary to help the model taking care of the longer term maturities, since the more negative the H , the faster the ATM skew decay for larger T .

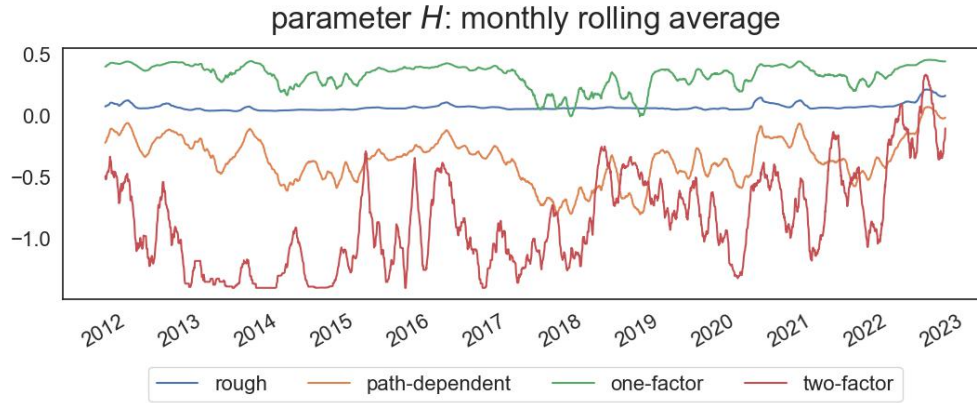


Figure 8: Evolution of monthly average of calibrated parameter H of different Bergomi models calibrated to maturities between one week to three years.

4.3 What does negative H mean?

On the short end, the large negative values of H can be difficult to interpret. For small maturities, the large negative calibrated values of H observed in the path-dependent and one-factor model on Figure 4 can be attributed to a highly erratic/spiky behavior of the volatility process, see Appendix C, which might not be well explained by a rough process, but may indicate the presence of jumps. Indeed, negative $H \leq -1/2$ can be related to jump regimes at the limit when $\varepsilon \rightarrow 0$ under Heston's dynamics [2]. These negative values for H also appeared in Figure 3 and Appendix B of [7] for the problem of joint calibration of SPX and VIX. When calibrated on longer maturities, the path-dependent model still displays a negative calibrated H as shown in Figure 8, although less extreme. This can be more easily explained by looking at the ATM skew as follows.

In two recent independent studies [22, 38], two different power laws have been used to fit the short and long end of the SPX ATM skew. We perform the same experiment by fitting two power laws both in the form of $cT^{H-1/2}$ on the average SPX ATM skew using linear regression. Specifically, we fit one power law on $(\log T, \log \bar{S}_T^{spx})$, with $\bar{S}_T^{spx}$ the average SPX ATM skew over the period for $T < \tau$ and another power law for $T \geq \tau$ and infer the two different value of H at different timescales. In Figure 9, τ is chosen to be 4 months based on the highest average R^2 value of the two linear regressions among all possible values of τ between one week to three years:

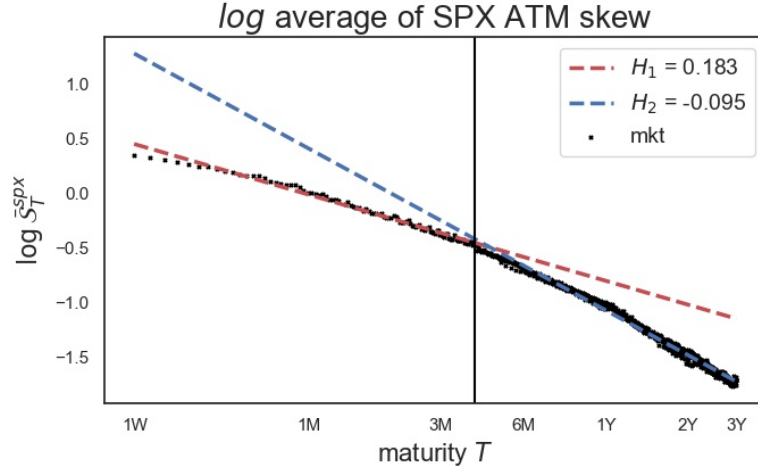


Figure 9: Fitting the $\log \bar{S}_T^{spx}$ with two linear regressions on $\log T$, with the first linear regression performed on $T \in [1W, 4M)$ and the second linear regression performed on $T \in [4M, 3Y]$. The slope of each linear regression represents estimated H at different timescales.

The decent fit of the blue dash line in Figure 9 suggests that long term ATM skew can be well-captured by a power law but with a negative $H_2 = -0.095$. We now show what happens to the estimation of H_2 by moving the cut-off time τ between one month to one year:

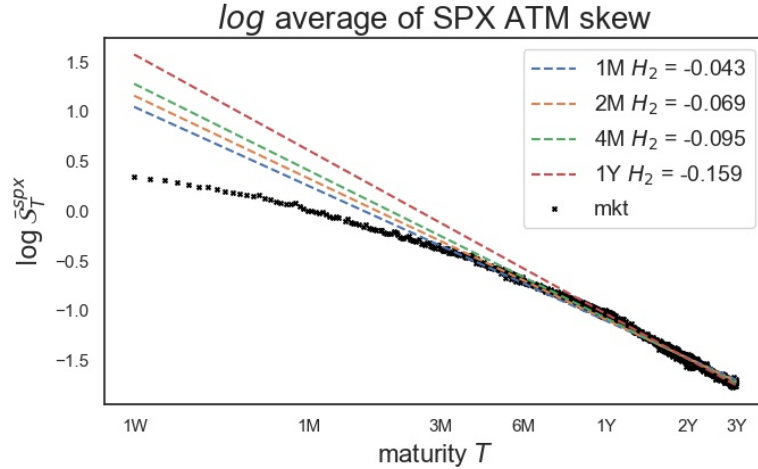


Figure 10: Fitting the $\log \bar{S}_T^{spx}$ with linear regression on $\log T$ for $T \in [\tau, 3Y]$ with $\tau \in \{1M, 2M, 4M, 1Y\}$. The slope of each linear regression represents estimated H_2 at different timescales.

The estimated H_2 becomes more negative as τ increases. This suggests that on average, the SPX ATM skew decays even faster than previously reported in [13, 28, 31, 32], where it is believed the decay coefficient is greater than $-1/2$, i.e. $H \geq 0$. The restriction $H > 0$ in the rough Bergomi model makes it impossible to match the average speed of SPX ATM skew decay for large T and partially explains its poor fit to the long end of the term structure of the SPX ATM skew in Figure 6. Similar negative values of H_2 have also appeared in Figure 4.8 of [37] when fitting the ATM skew on a specific day; and in Appendix C of [22] for fitting the average log ATM skew of a different and narrower time window.

To further validate the negative H_2 observed in Figures 9 and 10, we fit the daily SPX log ATM skew term structure against $\log T$ for $T \in [1Y, 3Y]$ via linear regression and plot the evolution of H_2 in Figure 11 with calibrated H from the rough and path-dependent Bergomi models (calibrated to the ATM skew for maturities between one week to three years) added in for comparison:

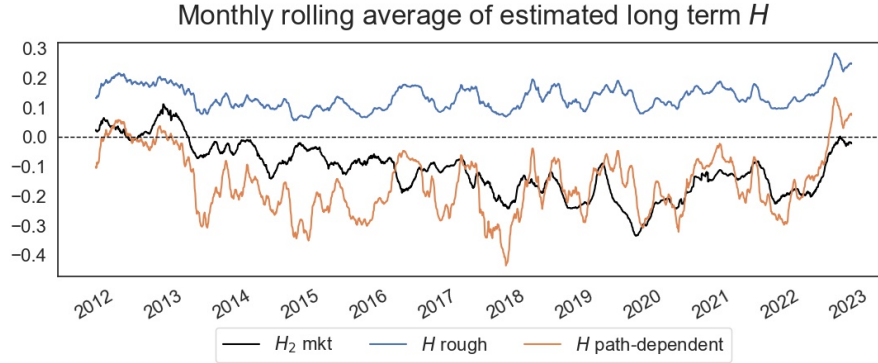


Figure 11: Evolution of market H_2 estimated by fitting the daily log SPX ATM skew against $\log T$ for $T \in [1Y, 3Y]$ compared to calibrated value of H of the rough and path-dependent Bergomi models (calibrated to the ATM skew for maturities between one week to three months). The dotted line represents the value zero.

Figure 11 shows that the estimated market H_2 is almost always negative apart from 2012 to mid 2013, thus providing further support to the case of negative H for SPX long term ATM skew decay.

Why should one care about the precise long term decay of the ATM skew? The long term ATM skew decay is linked to dynamical properties of the model, for instance via the skew stickiness ratio (SSR), where $\mathcal{S}_T \propto T^{SSR-2}$ for large T [13]. Estimating the SSR could then help in deducing H_2 through the relation $H_2 \approx SSR - 1.5$. In particular, the restriction of $H \in (0, 1/2)$ in rough volatility models leads to an $SSR \in (1.5, 2)$. We performed a first estimation of SSR using options with one year maturity in our data. The resulting SSR estimation fluctuates too much between 0.6 and 1.6 to obtain a reliable estimate of H_2 . However, the range of estimated SSR does suggest that H_2 can be negative which is inconsistent with rough volatility models, but can be better captured by the path-dependent and two factor models. In the next section, we will evaluate the dynamical properties of the models using a different experiment.

5 Empirical results: the dynamical comparison

5.1 Predicting Future SPX Implied Volatility Surface

We now compare the dynamical performance of each model by looking at how well they predict the future SPX volatility surface as described in Section 3.2.3. That is, for a given model, we fix the daily calibrated parameters obtained while fitting the global SPX smile in Section 3.2.1. We keep the calibrated parameters fixed for the next 20 working days, while taking in the daily market forward variance curve $\xi_0(\cdot)$ and see how well the model predicts SPX volatility surface in the next 20 working days, measured by the RMSE error between model vs. market mid data.

The box plot in Figure 12 shows the distribution of RMSE of prediction quality as across the whole period when the parameters are calibrated only to short maturities (one week to three months). The path-dependent Bergomi model scores the best performance. The one-factor Bergomi model

performs better compared to the rough Bergomi model for the first six forward business days and shares similar performance afterwards.

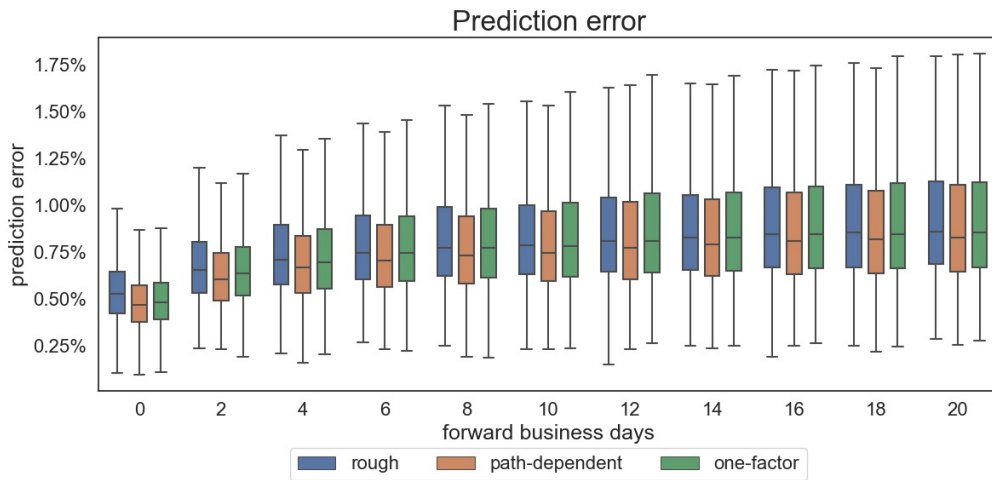


Figure 12: Box plot distribution of prediction RMSE between different Bergomi models across the entire period for maturities between one week to three months. Different values on the box plot represent 25, 50 and 75 quantiles, while the whiskers are calculated as 1.5 multiplied by the inter-quartile range away from 25 and 75 quantiles.

The box plot in Figure 13 shows the distribution of RMSE of prediction quality as across the whole period when the parameters are calibrated to longer maturities (one week to three years). The two-factor Bergomi model is the best among all the models in predicting future volatility surface, followed by the path-dependent Bergomi model. The rough Bergomi model slightly outperforms the one-factor model, however is considerably inadequate when compared to the other models.

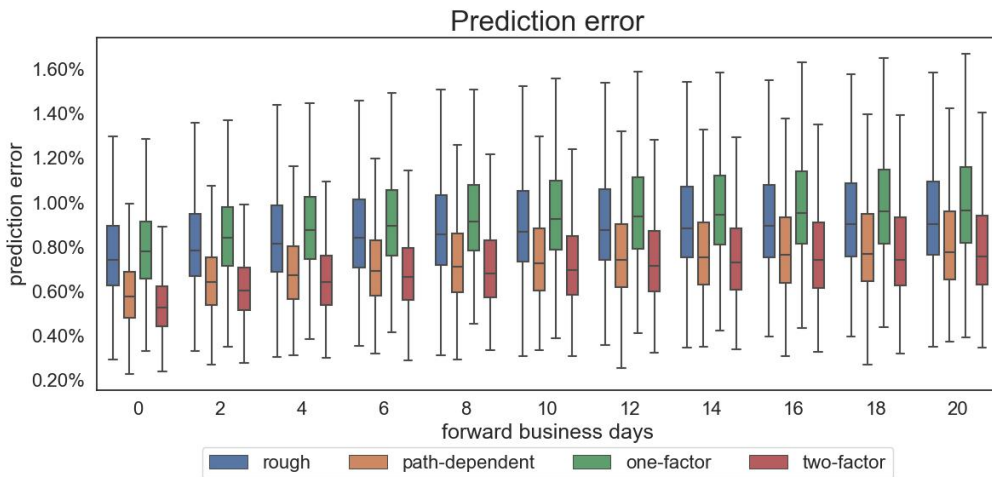


Figure 13: Box plot distribution of prediction RMSE between different Bergomi models across the entire period for maturities between one week to three years. Different values on the box plot represent 25, 50 and 75 quantiles, while the whiskers are calculated as 1.5 multiplied by the inter-quartile range away from 25 and 75 quantiles.

Both graphs show that the dynamical evolution of rough volatility model is far less consistent

with market data than its path-dependent and Markovian-counterparts. This further highlights the rigid structural issues associated with rough volatility models.

5.2 Spurious Roughness

We now use the same methodology in [33] to estimate the statistical roughness of the instantaneous volatility process via its proxy, the realized volatility process RV of all Bergomi models, as instantaneous volatility is unobservable in practice. First, we simulate the trajectories of the instantaneous volatility $\sqrt{V}$ and $\log S$ under each model, with time step size of 5 minutes for $T = 10$ years. Next, we compute the estimated daily RV via the following:

$$RV = \sqrt{\sum_{i=1}^n \left(\log(S_{t_i}/S_{t_{i-1}}) \right)^2},$$

with $t_n - t_0 = 1$ day and then calculate the empirical q -variation of the form:

$$m(q, \Delta) = \frac{1}{N} \sum_{k=1}^N \left| RV_{k\Delta} - RV_{(k-1)\Delta} \right|^q$$

for different values of q and timescale Δ . To remain as close as possible to the experiments performed in [33], we choose $q \in \{0.5, 1, 1.5, 2, 3\}$ and timescale $\Delta = 1, 2, \dots, 50$ Days. By fixing q , we compute the empirical q -variation and then use linear regression on $(\log \Delta, \log m(q, \Delta))$ to estimate its slope ζ_q . Next, we plot (q, ζ_q) and the slope of the graph (q, ζ_q) is the estimated Hurst index of the simulated process RV .

To simulate instantaneous volatility $\sqrt{V}$, we use the calibrated parameters of each model for the implied volatility surface on the day 23 October 2017, calibrated for maturities up to three months with the same seed. For this date, the calibrated parameters for the models are:

Model	Calibrated parameters
rough	$\eta = 1.28, \rho = -0.9397, H = 0.0787$
path-dependent	$\eta = 0.02551, \rho = -0.6884, H = -1.2761$
one-factor	$\eta = 0.7555, \rho = -0.6842, H = -0.3639$
two-factor	$\eta = 0.4297, \eta_\ell = 0.9843, \rho = -0.6849, H = -0.4973$

Table 5: Calibrated model parameters of each model for the implied volatility surface on 23 October 2017, with calibrated maturities between one week and three years.

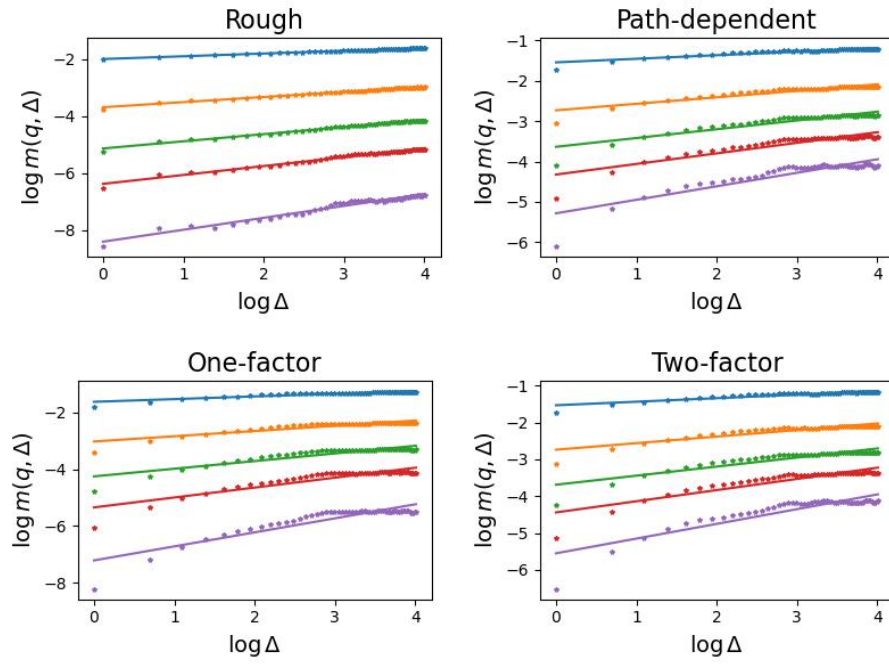


Figure 14: The log-log plot of different q -variations of a sample path of the realized volatility process of different models.

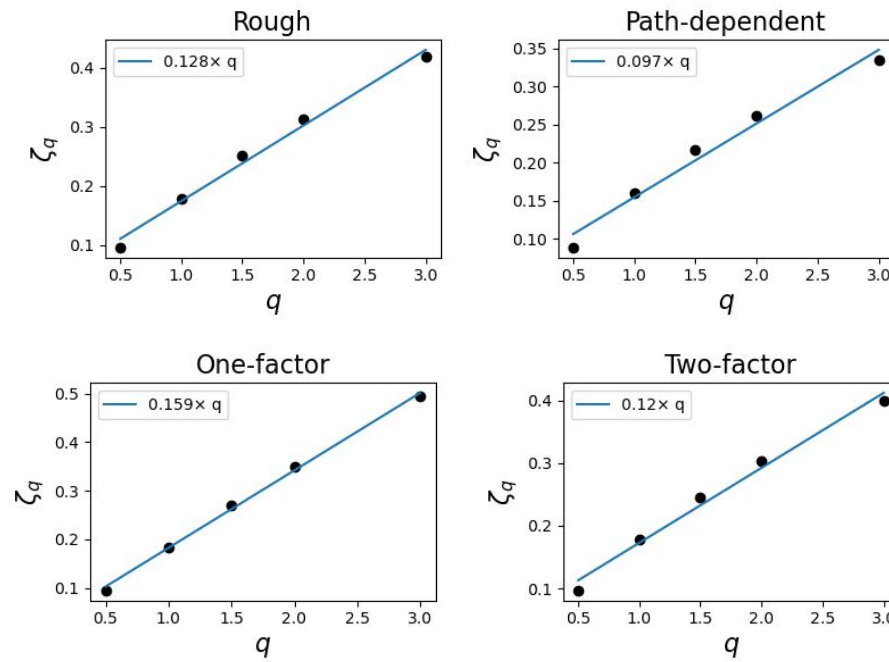


Figure 15: Plot of ζ_q against q , with the slope being the estimated Hurst index of different models.

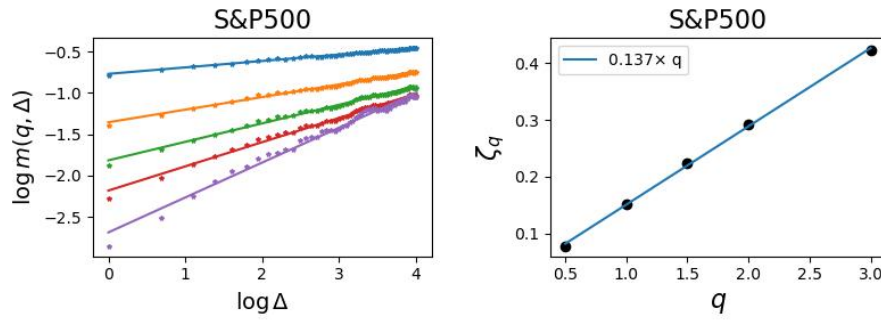


Figure 16: LHS: The log-log plot of different q -variations of the realized volatility time series of S&P500 between 2007 to 2017. RHS: Plot of ζ_q against q , with the slope being the estimated Hurst index of the S&P500 realized volatility time series.

From Figure 14 and 15, we see that the estimate for the Hurst index of the rough Bergomi model is $\hat{H} \approx 0.128$ comparing to the calibrated value of $H = 0.0787$ for the instantaneous volatility. For the other models, the estimated Hurst index $\hat{H}$ of RV returns a value between 0.097 to 0.159, comparing to the theoretical Hurst index of 0.5 of their instantaneous volatility. Figure 16 shows the estimated $\hat{H} \approx 0.137$ of the actual S&P500 realized volatility over a ten year period between 2007 to 2017, with data coming from the Oxford-Man Institute of Quantitative Finance Realized Library for comparison.

The sample paths of realized volatility used for the estimation of H in Figure 14 and 15 of each Bergomi model are included in Appendix C alongside with the annualized realized volatility time series of the S&P500. These sample paths all seem to have an erratic and spiky behavior. Combined with the small estimated statistical $\hat{H}$, this suggests that the rougher appearance of the realized volatility process does not mean that the underlying process has to be rough, and the estimation of H is not enough to discriminate between models, since all of our non-rough models produce similar spurious roughness on any realistic timescales, in line with [1, 19, 39, 49].

A Sample fits of SPX smile: one week to three months

July 03, 2013

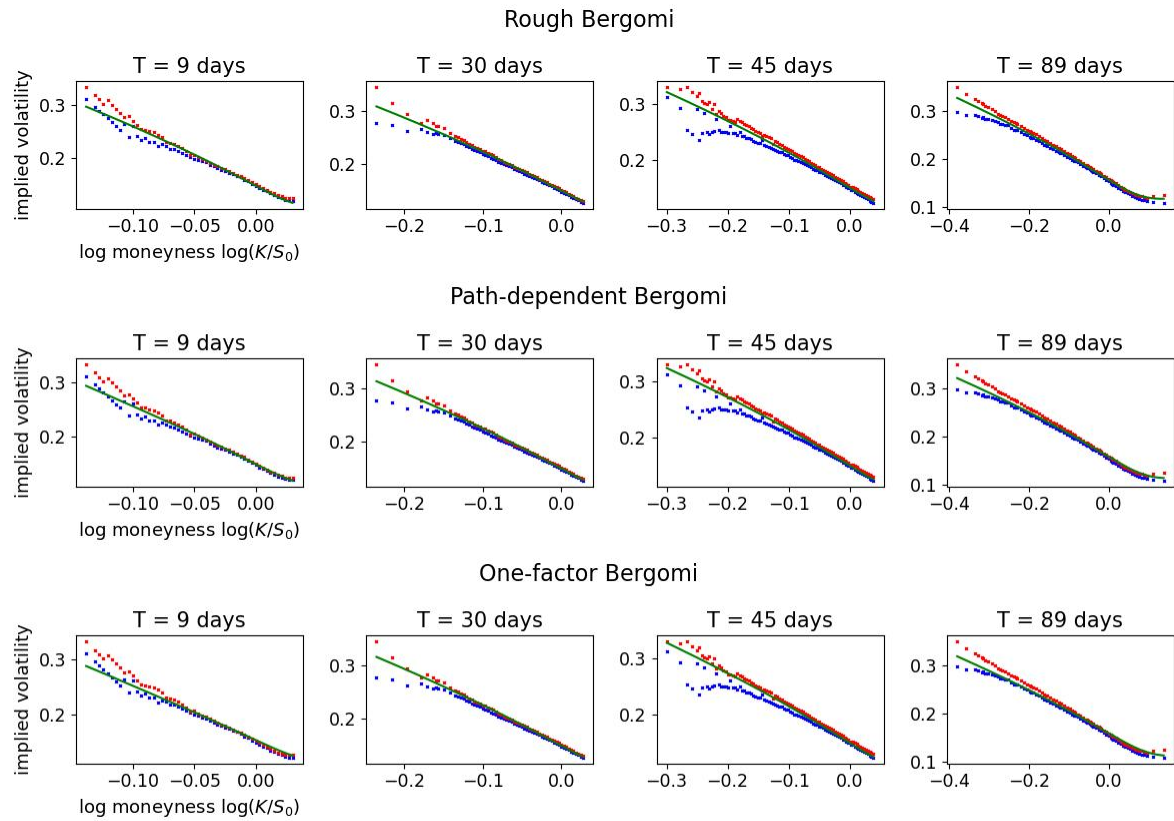


Figure 17: SPX smiles (bid/ask in blue/red) on 3 July 2013 calibrated by different Bergomi models (green lines).

October 23, 2017

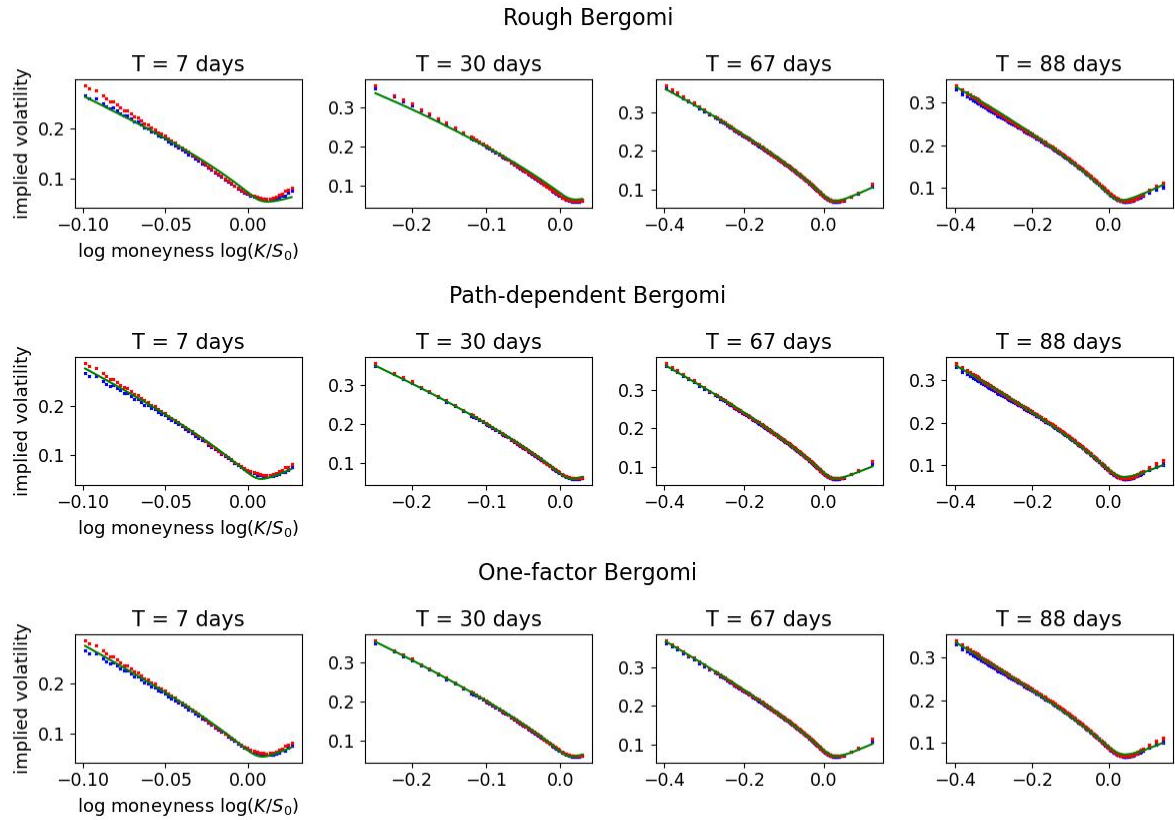


Figure 18: SPX smiles (bid/ask in blue/red) on 23 October 2017 calibrated by different Bergomi models (green lines).

B Sample fits of SPX smile: one week to three years

July 03, 2013

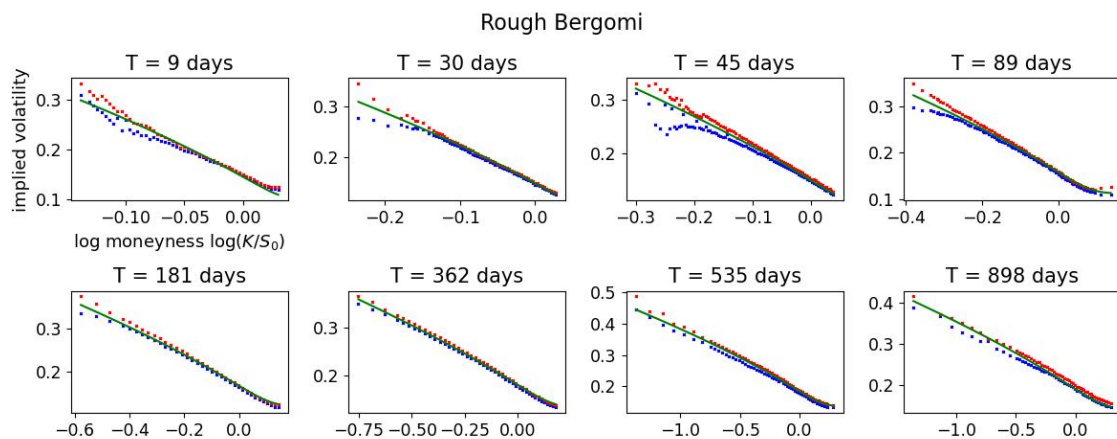


Figure 19: SPX smiles (bid/ask in blue/red) on 3 July 2013 calibrated by **rough** Bergomi models (green lines).

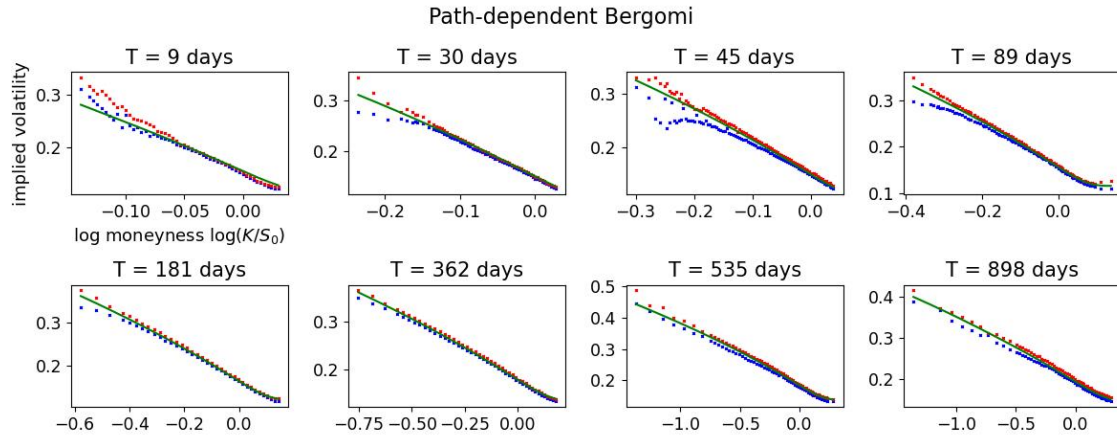


Figure 20: SPX smiles (bid/ask in blue/red) on 3 July 2013 calibrated by **path-dependent** Bergomi models (green lines).

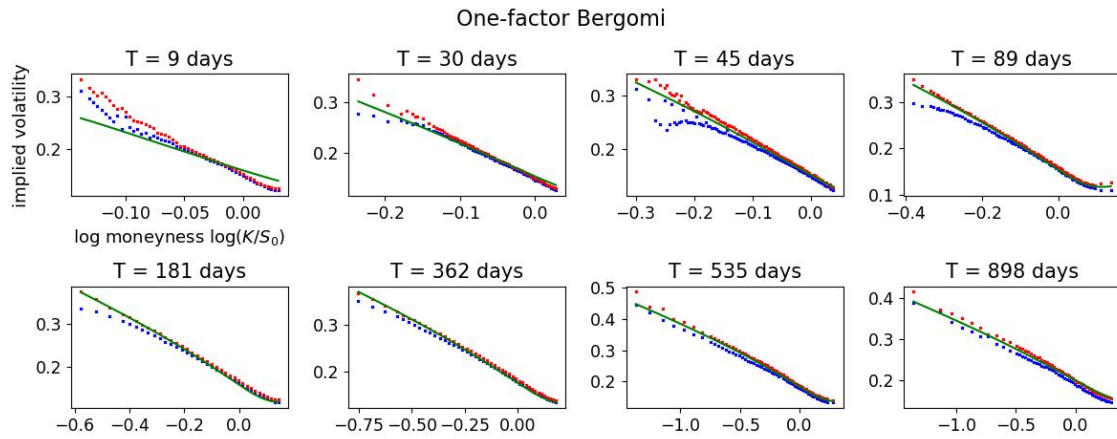


Figure 21: SPX smiles (bid/ask in blue/red) on 3 July 2013 calibrated by **one-factor** Bergomi models (green lines).

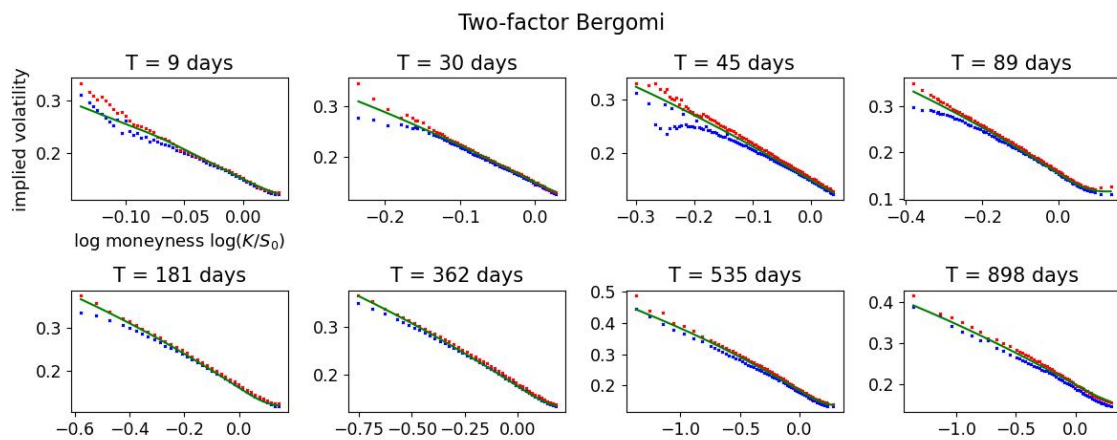


Figure 22: SPX smiles (bid/ask in blue/red) on 3 July 2013 calibrated by **two-factor** Bergomi models (green lines).

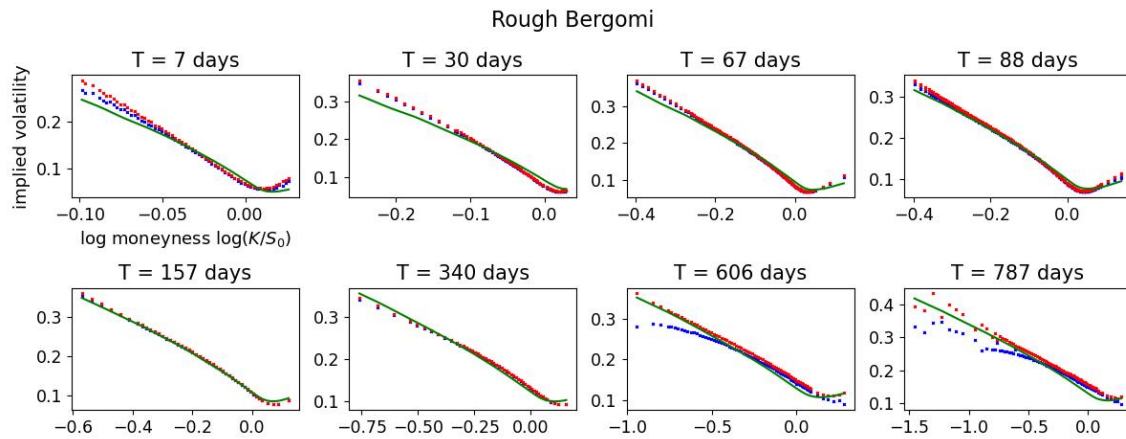


Figure 23: SPX smiles (bid/ask in blue/red) on 23 October 2017 calibrated by **rough** Bergomi models (green lines).

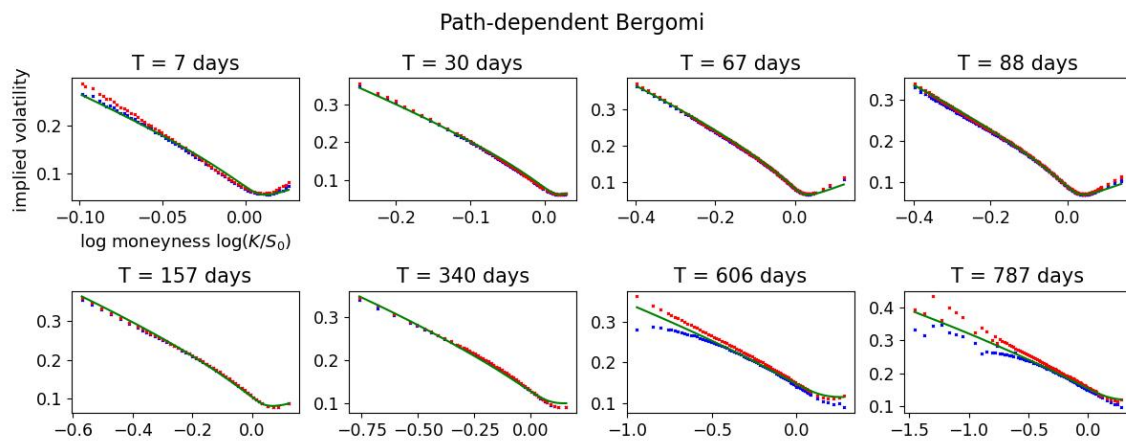


Figure 24: SPX smiles (bid/ask in blue/red) on 23 October 2017 calibrated by **path-dependent** Bergomi models (green lines).

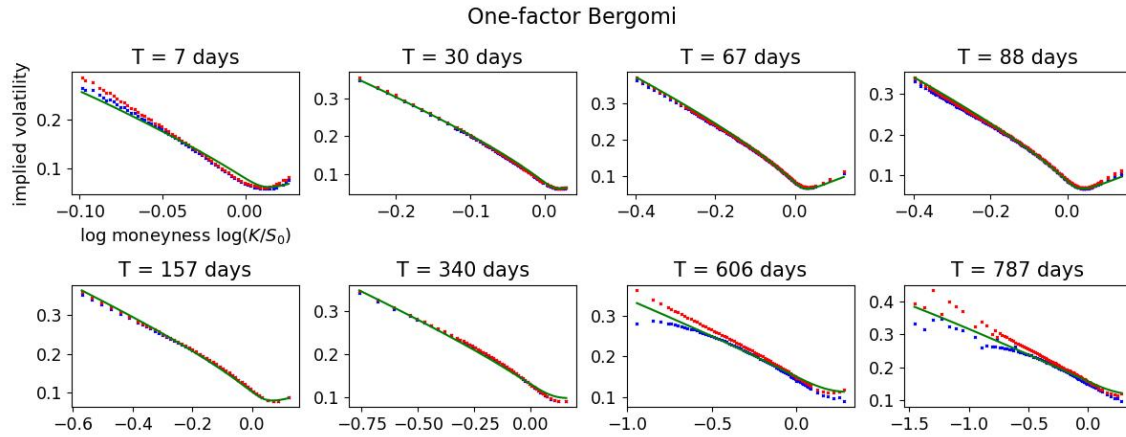


Figure 25: SPX smiles (bid/ask in blue/red) on 23 October 2017 calibrated by **one-factor** Bergomi models (green lines).

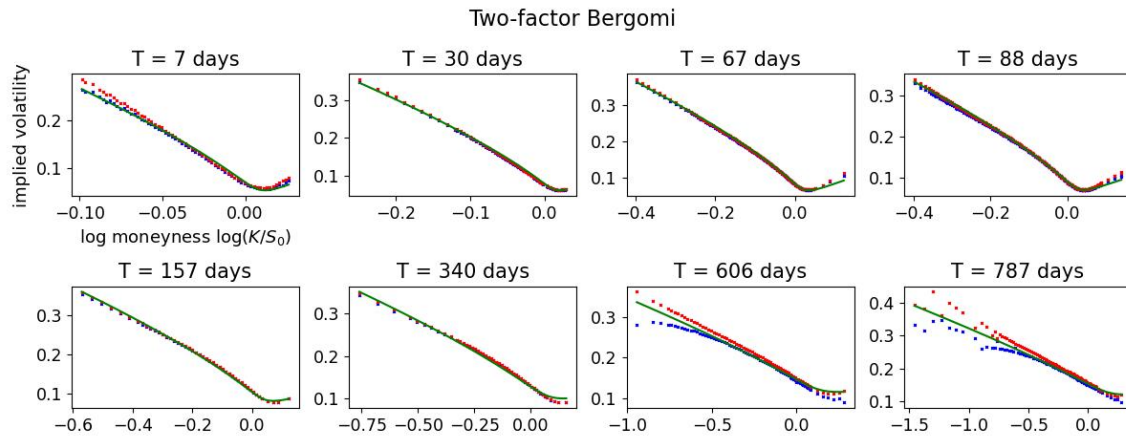


Figure 26: SPX smiles (bid/ask in blue/red) on 23 October 2017 calibrated by **two-factor** Bergomi models (green lines).

C Simulated realized volatility

Figure 27 shows the simulated trajectories of the realized volatility process under each Bergomi model with the same seed, using the calibrated parameters from Table 5 based on the implied volatility surface on 27 October 2017, and the annualized realized volatility time series of S&P 500 between 2007 to 2017 with data coming from the Oxford-Man Institute's Realized Library.

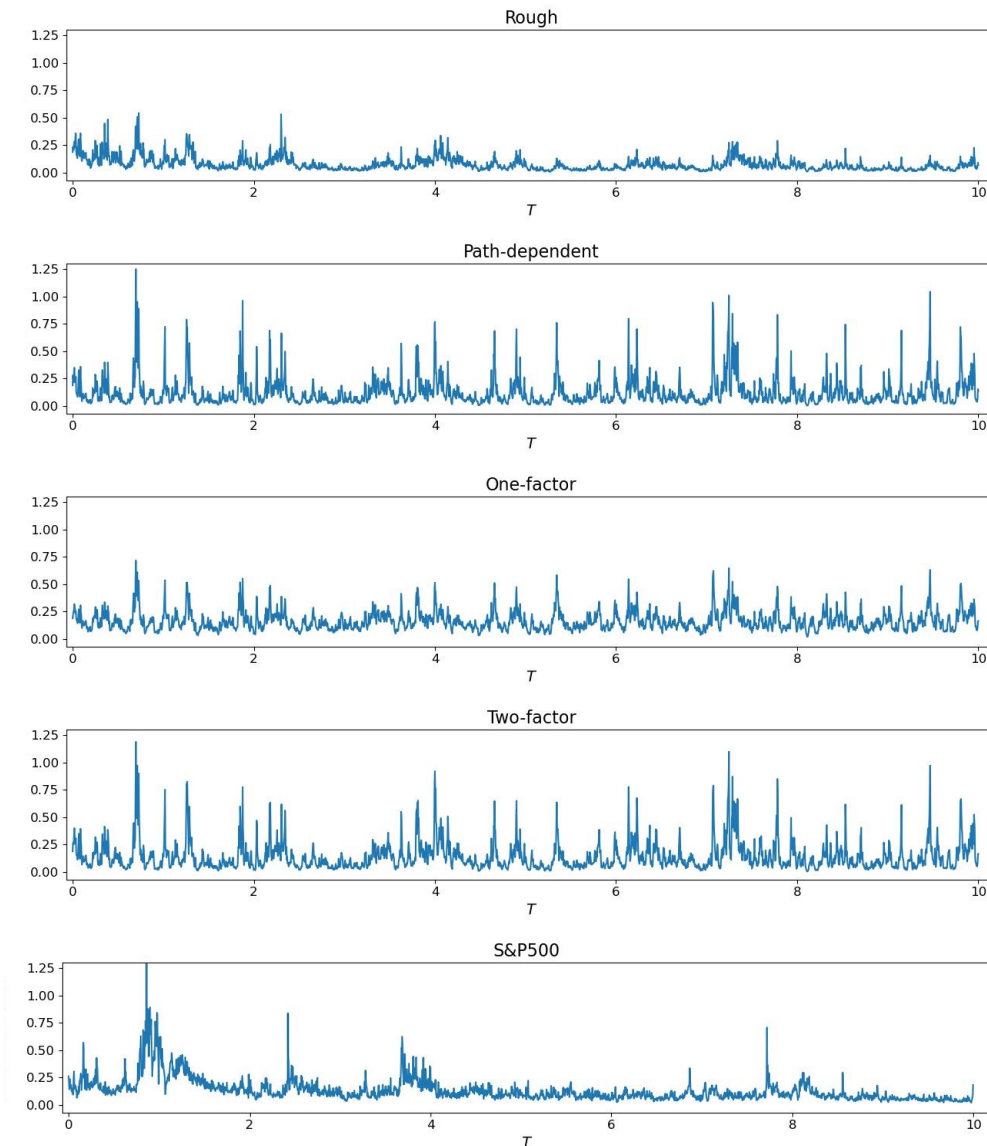


Figure 27: Simulated trajectory of the realized volatility process RV of different models with the same seed using the calibrated parameters of the implied volatility surface dated 27 October 2017 and the annualized realized volatility time series of S&P 500 between 2007 to 2017.

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